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Temperature-dependent upconversion luminescence and dynamics of NaYF₄:Yb³⁺/Er³⁺ nanocrystals: influence of particle size and crystalline phase†

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Oleic acid-capped NaYF₄:Yb³⁺/Er³⁺ upconversion nanocrystals (UCNCs) with different sizes and crystalline phases were prepared, and their temperature-dependent upconversion luminescence (UCL) and dynamics were studied. It is interesting to observe that the temperature-dependent behavior of UCL for the β -phase (25 nm, 45 nm and bulk) and α -phase (<10 nm) UCNCs is quite different. The UCL intensity of Er³⁺ ions of the β -phase NaYF₄ demonstrates a maximum around 100 K, while the intensity of the α -phase quenches monotonously with elevated temperature (10–400 K). The intensity ratio of ²H_{11/2}–⁴I_{15/2} to ⁴S_{3/2}–⁴I_{15/2}, *R*_{HS}, increases solely with temperature for β -phase NaYF₄, while for the α -phase sample, it demonstrates a complex and indeterminate variation as a function of temperature. In the β -phase samples, rising processes were observed in the dynamics of Er³⁺ ions, while in the α -phase sample, no rising process was observed and the decay processes of Er³⁺ ions were bi-exponential. The rationale for these different temperature-dependent UCL properties was explained carefully.

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1. Introduction

Upconversion (UC) phosphors, which can convert a low frequency excitation photon into a high frequency emission photon through two-photon or multi-photon processes, have attracted great interest in recent decades owing to their unique photoluminescent features and a large number of application potentials, such as background lighting sources, solid-state lasers, infrared detection, solar cell conversion, *etc.*^{1–6} In particular, some rare earth (RE) doped fluoride UCNCs are attracting extensive current interest because of their relatively high UC efficiency and significant application potential in the field of *in vivo* imaging, infrared (IR) photo-dynamic therapy, bio-labels and bio-sensors.^{7–9} Among various fluoride UC phosphors, the micro-sized polycrystalline powder of β -phase NaYF₄:Yb, Er is considered as one of the most efficient under the excitation of a 980 nm laser diode, however, its UC

efficiency is only 3%, and that of the corresponding nanosized phosphors (~10 nm) is only one of tenth of the micro-sized form.^{10,11} Therefore, it is important to further improve the efficiency of the fluoride UC phosphors and to search for novel efficient UC phosphors.

It is known that the UCL strongly depends on dopant concentrations, excitation power, crystallite phases and particle sizes, and thermal conditions.^{12–15} Actually, there are two types of crystallite phases in NaYF₄ at ambient pressure and temperature: the cubic phase (α -phase) and the hexagonal phase (β -phase). It was reported that the UC intensity of the green emission of β -NaYF₄:Yb, Er is 10 times stronger than that of α -NaYF₄:Yb, Er and the overall (green plus red) emission of the former is 4.4 times higher than that of the latter.^{16–18} Generally, with the decrease of particle size, the UC strength was suppressed due to the improved surface adsorption of large phonon bonds, such as water and the unexpected crystallinity of smaller nanoparticles (NPs), and, the ratio of the green to the red was also considerably modulated.^{19,20} Despite a number of studies related to NaYF₄:Yb, Er UCNCs having been performed, some significant problems have not been completely understood. For instance, essentially, how the change of crystalline phase as well as particle size influences the UCL of the phosphors, including radiative transition, non-radiative relaxation and ET processes.

The studies on temperature-dependent UCL and dynamics of NaYF₄:Yb, Er UCNCs with different particle sizes and

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† Electronic supplementary information (ESI) available: UCL decay time of the transitions at 416 nm, 544 nm and 656 nm of Er³⁺ ions in 25 nm α -NaYF₄ Fig. S1(a)–(c); bulk β -NaYF₄ Fig. S2(a)–(c) for samples with different temperature. From the best fitting functions it can be seen that for the 25 nm and bulk samples, their luminescent dynamics are similar to those of the 45 nm samples in our work. See DOI: 10.1039/c3dt53562b

crystalline phases are significant for understanding these questions. Temperature has two effects on the UCL of nanomaterials. On the one hand, it can generate heat radiation and stimulated luminescence, on the other hand, it promotes the nonradiative transitions and quenching.^{20–22} The study of their variable temperature spectra will help to further understand UC mechanisms and the use of the materials for potential thermal sensing and biosensor applications.^{21,23} It was reported²⁴ that the luminescent intensities of Er^{3+} and Yb^{3+} codoped Y_2O_3 nanocrystals decreased with the increasing temperature from 10 K to 300 K. This effect was explained by the enhancement of nonradiative multiphonon relaxation rates, which reduces the radiative transition probability at high temperatures. Van der Ziel *et al.*²⁵ found that the intensities of YF_3 increased and then decreased with the temperature raising up to room temperature. The depopulation of the higher sub-levels of the $^4\text{I}_{11/2}$ at lower temperatures and multiphonon de-excitation at higher temperatures were proposed as dominant factors that govern the temperature-dependent behavior. Shan *et al.*²⁶ reported UCL of $\text{NaYF}_4:\text{Yb}^{3+}, \text{Er}^{3+}$ UCNCs at various temperatures from 300 K to 450 K and observed a similar phenomenon. So far, there is no complete work related to temperature-dependent UCL and dynamics of $\text{NaYF}_4:\text{Yb}, \text{Er}$ with different sizes and crystalline phases.

In the present work, we prepared oleic acid (OA)-capped $\text{NaYF}_4:\text{Yb}^{3+}, \text{Er}^{3+}$ NCs with different sizes and phases and investigated their temperature-dependent UCL and dynamics systematically. Some interesting temperature-dependent UCL properties were observed, which depended on both the crystalline phase and particle size of $\text{NaYF}_4:\text{Yb}^{3+}, \text{Er}^{3+}$ UCNCs.

2. Experimental section

2.1. Materials

All of the chemicals were of analytical grade and were used without further purification. NaOH , NH_4F , methanol, and cyclohexane were purchased from Sinopharm Chemical Reagent Co., Ltd. Octadecene was purchased from Sigma (America). OA (>90%) was obtained from Alfa Aesar. Rare earth (RE) chlorides (LnCl_3 , Ln: Y, Yb, Er) (>99.99%) were purchased from Baotou Ruike Functional Materials Co. Ltd.

2.2. Synthesis of $\text{NaYF}_4:\text{Yb}, \text{Er}$ UCNCs

The OA-capped $\text{NaYF}_4:18\% \text{ Yb}, 2\% \text{ Er}$ NCs were synthesized following a procedure reported previously.²⁷ Firstly, YCl_3 (0.2416 g), YbCl_3 (0.0697 g) and ErCl_3 (0.0077 g) were mixed with 6 ml OA and 15 ml octadecene (ODE) and heated to 160 °C to form a homogeneous solution, and then cooled down to room temperature. Then, 10 ml methanol solution containing NaOH (0.1 g) and NH_4F (0.148 g) was slowly added into the solution with magnetic stirring, then the solution was stirred continuously for 30 min. Subsequently, the solution was heated to 120 °C to form a pellucid solution and removal of residual water and oxygen, then heated to 280, 300, or 320 °C and maintained for 1 h under nitrogen atmosphere to

prepare NCs with different sizes. After the solution was cooled down naturally, NCs were precipitated from the solution with ethanol and washed with cyclohexane for three times. The bulk material was obtained by annealing the 320 °C sample at 600 °C for 3 hours.

2.3. Characterization

The size and morphology of as-prepared UCNCs were recorded on a Hitachi H-8100IV transmission electron microscope (TEM) under an acceleration voltage of 200 kV. The as-prepared samples were dispersed in cyclohexane and dripped onto a copper grid for the TEM tests. The crystalline structure and purity of the as-prepared samples were characterized by X-ray power diffraction (XRD) with a Rigaku D/max 2550 X-ray diffractometer, using a monochromatized Cu target radiation source ($\lambda = 1.54 \text{ \AA}$). The UCL spectra were measured by an Andor Shamrock SR-750 fluorescence spectrometer. A CCD detector combined with a monochromator was used for signal collection from 300 nm to 850 nm. A continuous 980 nm diode laser was used to pump the samples for the measurement of steady-state emission spectra. In the measurement of UC dynamics, a tunable Nd:YAG laser pumped OPO laser was used as the excitation source, with a pulse duration of 10 ns, repetition frequency of 10 Hz and line width of $4\text{--}7 \text{ cm}^{-1}$ and was fixed at 980 nm. All measurements were tested under vacuum, and the temperature was changed from 10 K to 400 K.

3. Results and discussion

3.1 Structure and morphology of the $\text{NaYF}_4:\text{Yb}, \text{Er}$ UCNCs

The transmission electron microscopy (TEM) images of as-prepared NaYF_4 prepared at different temperatures are shown in Fig. 1(a–c), with a scanning electron microscope (SEM) image in Fig. 1(d). A histogram of the particle size distribution was obtained by direct observation on TEM bright field images, as shown in Fig. 1(e–g). It can be seen that when the reaction temperature was at 280 °C, small NaYF_4 UPNCs were formed and they showed a rather wide size distribution. The mean particle size was determined to be $6.5 \pm 6.1 \text{ nm}$. From the high-resolution patterns, a fringe with lattice distance of 0.3 nm was observed, corresponding to the (111) face of cubic NaYF_4 , implying the formation of a crystalline phase. However, it is also clear that some regions contained an amorphous phase. As the temperature increased further, the average sizes of NaYF_4 UCNCs increased and became homogeneous, which were $25 \pm 1.5 \text{ nm}$ when prepared at 300 °C and $45 \pm 1.9 \text{ nm}$ at 320 °C. For the annealed bulk sample (see Fig. 1d), the shape of particles is irregular due to the accumulation of small NPs, and the size of the particles ranges from several hundreds of nanometers to several micrometers. Fig. 1h shows the XRD patterns of the as-prepared NaYF_4 UCNCs fabricated at different temperatures in comparison to the standard cards of $\alpha\text{-NaYF}_4$ and $\beta\text{-NaYF}_4$. It can be distinguished that the samples prepared at 280 °C demonstrate the cubic phase (JCPDS: 077-2042). As the reaction temperature increases to 300 °C, NaYF_4

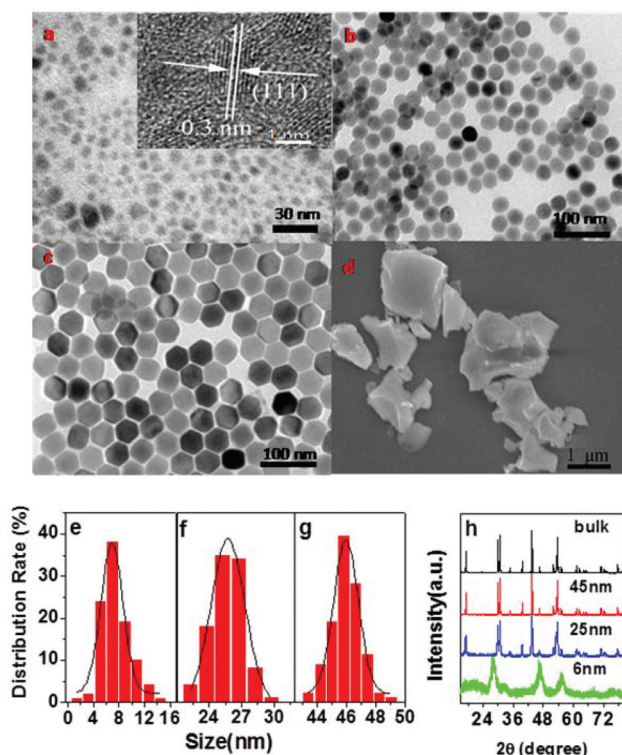


Fig. 1 TEM images (a–c), SEM image (d), size distribution (e–g) and XRD (h) patterns of $\text{NaYF}_4\text{:Yb}^{3+}, \text{Er}^{3+}$ nanocrystals at different reaction temperatures of 280 °C (a, e cubic phase, JCPDS: 077-2042), 300 °C (b, f hexagonal phase), and 320 °C (c, g hexagonal phase) and bulk (d, hexagonal phase).

has transformed into β -phase. At higher temperature, the pure hexagonal phase also appears (JCPDS standard card no. 28-1192).

3.2 Power-dependence of $\text{NaYF}_4\text{:Yb, Er}$ UCNCs

The normalized UCL spectra of $\text{NaYF}_4\text{:Yb, Er}$ UCNCs with different sizes under the excitation of a 980 nm laser diode are shown in Fig. 2a. In all the emission spectra of $\text{NaYF}_4\text{:Yb, Er}$,

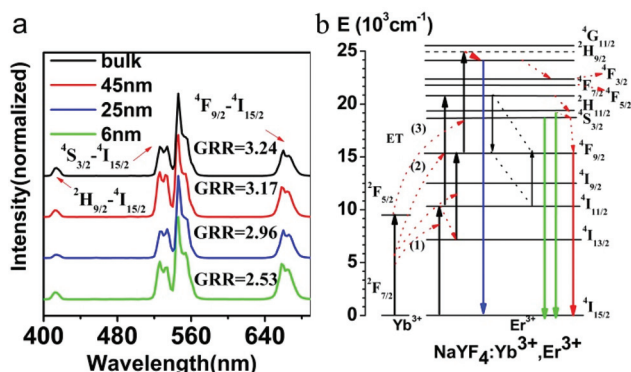


Fig. 2 (a) The UCL spectra of $\text{NaYF}_4\text{:Yb}^{3+}, \text{Er}^{3+}$ with different sizes excited by a 980 nm laser diode of 1.8 W mm^{-2} . (b) The schematic of the luminescence processes of $\text{NaYF}_4\text{:Yb}^{3+}, \text{Er}^{3+}$ under the excitation of 980 nm light. The solid lines present radiative transitions and the dashed lines are nonradiative ones.

Er UCNCs, the blue ${}^2\text{H}_{9/2}\text{--}{}^4\text{I}_{15/2}$, green ${}^4\text{S}_{3/2}, {}^2\text{H}_{11/2}\text{--}{}^4\text{I}_{15/2}$ and red ${}^4\text{F}_{9/2}\text{--}{}^4\text{I}_{15/2}$ transitions can be clearly identified. The green emissions of ${}^4\text{S}_{3/2}, {}^2\text{H}_{11/2}\text{--}{}^4\text{I}_{15/2}$ are dominant for all the samples. It can be seen that the intensity ratio of the red to green slightly increases with the decreasing particle size. It should be noted that the overall emission intensity of the samples increases considerably with the increasing particle size, as reported by the previous literature. Previously, we systematically studied the size-dependent UCL behavior of $\text{NaLuF}_4\text{:Yb, Er}$ with different phases, α -phase and β -phase, and observed that they demonstrated opposite results in that for the β -phase NaYF_4 , the intensity ratio of red to green slightly increases with the decreasing particle size, while on the contrary, in the α -phase sample, the intensity ratio of red to green decreases remarkably with the decreasing particle size. The difference of size-dependent UCL between α -phase and β -phase samples was attributed to the contribution of cross-relaxation processes.³⁰ In the α -phase samples, the cross-relaxation process happened easily due to the random distribution of rare earth ions, while in the β -phase samples, the cross-relaxation process hardly occurred due to the strict distribution of rare earth ions.³⁰

Fig. 2b shows the schematic of the UC excitation pathways and the emission processes for the $\text{NaYF}_4\text{:Er}^{3+}, \text{Yb}^{3+}$ under 980 nm excitation.¹⁹ Firstly, Er^{3+} ions on ${}^4\text{I}_{15/2}$ are excited to ${}^4\text{I}_{11/2}$ by the ground state absorption (GSA) and/or energy transfer (ET) of $\text{Yb}^{3+}\text{--}\text{Er}^{3+}$. Then the same laser pumps the excited-state electrons from the ${}^4\text{I}_{11/2}$ to the ${}^2\text{H}_{11/2}$ levels *via* excited-state absorption (ESA) or the second-step ET of $\text{Yb}^{3+}\text{--}\text{Er}^{3+}$, the electrons relax to ${}^4\text{S}_{3/2}, {}^2\text{H}_{11/2}$, generating green ${}^4\text{S}_{3/2}, {}^2\text{H}_{11/2}\text{--}{}^4\text{I}_{15/2}$ transitions. The populating pathway of the red ${}^4\text{F}_{9/2}$ level is as follows: after populating the ${}^4\text{I}_{11/2}$ level, the electrons further relax to ${}^4\text{I}_{13/2}$, then are excited into ${}^4\text{F}_{9/2}$, generating ${}^4\text{F}_{9/2}\text{--}{}^4\text{I}_{15/2}$ transitions. Besides, the nonradiative relaxation from the green ${}^4\text{S}_{3/2}, {}^2\text{H}_{11/2}$ level to ${}^4\text{F}_{9/2}$ is another pathway for the population of the red level. The populating of the blue level is a three-photon process: some of the electrons on the red ${}^4\text{F}_{9/2}$ are further excited into ${}^2\text{H}_{9/2}$ through the third-step ET of Yb^{3+} to Er^{3+} , and generating blue ${}^2\text{H}_{9/2}\text{--}{}^4\text{I}_{15/2}$ transitions. Generally speaking, as the particle size decreases, due to the increase of the ratio of surface to volume, a number of surface defects with available large vibrational modes such as CO_3^{2-} and OH^- are involved. Large vibrational modes results in the increase of nonradiative relaxation channels, ${}^4\text{I}_{11/2}\text{--}{}^4\text{I}_{13/2}$ and ${}^4\text{S}_{3/2}, {}^2\text{H}_{11/2}\text{--}{}^4\text{F}_{9/2}$, leading to the relative increase of the red transitions. It should be also noted that as the doping concentration is high enough, cross-relaxation of ${}^4\text{F}_{7/2} + {}^4\text{I}_{11/2} \rightarrow {}^4\text{F}_{9/2} + {}^4\text{F}_{9/2}$ also occurs significantly, leading to the relative increase of the red emissions.

To further reveal the UC mechanism, the blue (${}^2\text{H}_{9/2}\text{--}{}^4\text{I}_{15/2}$), green (${}^2\text{H}_{11/2}/{}^4\text{S}_{3/2}\text{--}{}^4\text{I}_{15/2}$) and red (${}^4\text{F}_{9/2}\text{--}{}^4\text{I}_{15/2}$) emissions were measured as a function of excitation power, as shown in Fig. 3. Following the UC mechanism, the visible emission intensity is proportional to a certain power (n) of the infrared excitation intensity. For the blue emission, the values of n were determined to be 1.43, 2.19, 1.29 and 1.12. For the green emission,

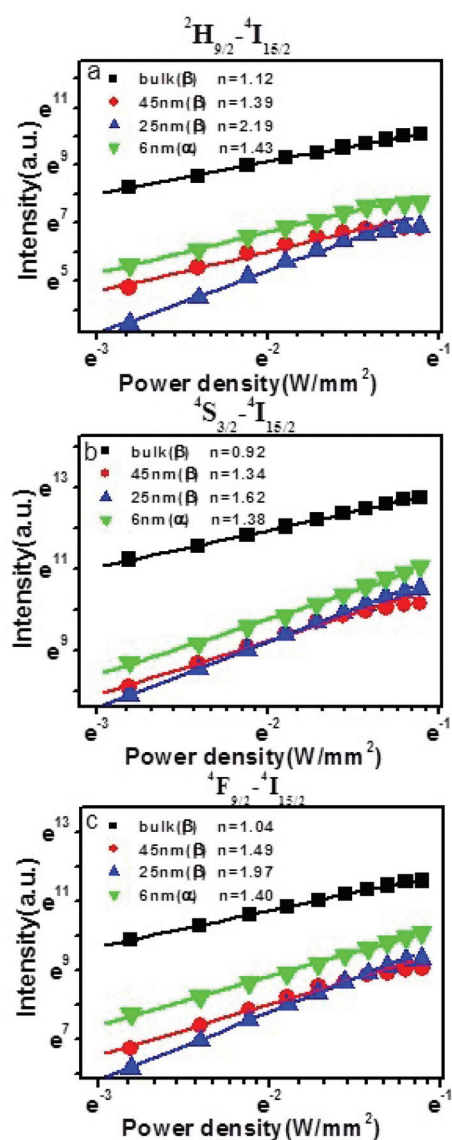


Fig. 3 Plot (ln–ln) of emission intensity versus excitation power in different sized particles. (a) Blue emission, (b) green emission and, (c) red emission.

the values of n were determined to be 1.38, 1.62, 1.34, and 1.92. For the red emission the values of n were determined to be 1.40, 1.97, 1.49 and 1.04. As is well known, for any up-conversion mechanism, the value of n presents the number of IR photons absorbed per visible photon emitted. For the two-photon processes (red and green emissions), n should equal or be close to 2 and for the three-photon processes (blue emission), n should be close to 3. However, in our experiment, the values of n were much smaller than the photon number for the transitions. This was mainly attributed to the competition between linear decay and UC processes for the depletion of the intermediate excited states, which was theoretically described by Pollnau *et al.*²² It can be deduced that when the particle size decreased from bulk to 25 nm, the value of n increased gradually, which was attributed to the suppression of the

saturation effect. As the particle size decreased, the linear decay process would increase due to the increase of non-radiative relaxation processes. Oppositely, when the particle size decreased from 25 nm to several nanometers, the value of n decreased, which could be attributed to greatly improved local thermal effect. For the 6 nm UCNCs, nonradiative transition process will increase significantly due to its worse crystallinity and improved surface adsorption. The local thermal effect will induce the temperature of the sample at the irradiated site to increase with the excitation power. Slight thermal quenching will cause the value of n to decrease, while strong thermal quenching will even lead to the decrease of UCL intensity with the excitation power.²⁸

3.3 Temperature-dependent UCL spectra of NaYF₄:Yb, Er NCs

The temperature-dependent UCL of NaYF₄:Yb³⁺, Er³⁺ NCs with different sizes were compared under 980 nm excitation, as shown in Fig. 4. It is interesting to observe that for the 6 nm α -phase sample, the intensity of UCL decreases solely with the increase of temperature for all the red, green and blue transitions, which can be attributed to significant thermal quenching of UCL due to its worse crystallinity and surface adsorption of water. For the other β -phase UCNCs (25 nm, 45 nm and bulk), the UCL intensity of Er³⁺ ions first increases with elevated temperature, and it demonstrates an optimum at a certain temperature, and then decreases with further increase of temperature, for all the transitions. For the bulk sample the optimum appears around 100 K, and for the 25 nm and 45 nm samples, the optimum appears around 150 K. Suyver *et al.*²⁹ pointed out that there is an energy gap between the two lowest crystal-field components $^2F_{5/2}|0\rangle$ and $^2F_{5/2}|1\rangle$ of the Yb³⁺ ions and the ET from $^2F_{5/2}|1\rangle$ to the lowest energy crystal-field component $^4I_{11/2}$ state of the Er³⁺ ions is very efficient. On one hand, with the increase of temperature, the relative population on $^2F_{5/2}|1\rangle$ of the Yb³⁺ ions will increase, leading to the increase of the excitation of Yb³⁺, as a consequence, the ET from Yb³⁺ to Er³⁺ will be more efficient, and the UCL intensity increases with the increase of temperature. On the other hand, the nonradiative relaxation processes will increase with the increase of temperature, leading to the quenching of UCL. These two competing processes lead the appearance of an optimum of UCL intensity at a certain temperature.

Fig. 5(a) shows the temperature-dependent UCL spectra of the 6 nm α -phase. For the 6 nm sample, the UCL of $^2H_{11/2}$, $^4S_{3/2}$ – $^4I_{15/2}$ and $^4F_{9/2}$ at 10 K has broader bands and the Stark splitting of the emission lines for different transitions is not obvious. This suggests that the local environments surrounding the Er³⁺ luminescent centers are very disordered and the inhomogeneous broadening of the emission lines is quite great. In the other words, the luminescent centers of Er³⁺ for the poorer-crystalline NaYF₄ contribute dominantly to the UCL at low temperature (10–50 K). Above 100 K, the broader emission bands nearly disappear and the emission lines with distinct Stark splitting can be clearly identified. The above result

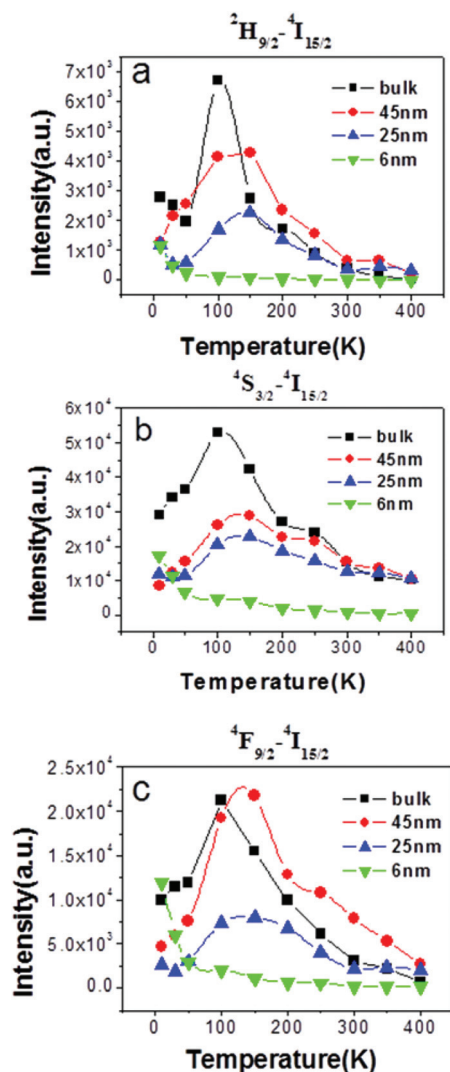


Fig. 4 Difference in up-conversion emission intensity of NaYF₄:Yb³⁺, Er³⁺ nanoparticles and bulk when taken as dry powder and excited under 980 nm LD excitation with the same power density (2.9 W mm⁻²) at verified temperatures from 10 K to 400 K in vacuum. Blue light (a), green light (b) and red light (c).

indicates that the luminescent centers surrounding the disordered local environments have completely quenched above 100 K due to possible ET from the centers to the defect states. It should be noted that previously we studied the frequency-selective excitation spectra of Eu³⁺ ions in some oxide compounds NCs such as Y₂O₃ and YVO₄ and identified different symmetry sites, inner sites and surface sites and observed that the emissions of the centers locating on the surface sites became broader, and the luminescent lifetime of the centers became shorter.^{30,31} From Fig. 5(a) it can be also seen that the overall intensity ratio of the red emission to the green emission ($R_{r/g}$), and the intensity ratio (R_{HS}) of $^2H_{11/2}-^4I_{15/2}$ to $^4S_{3/2}-^4I_{15/2}$ also varies significantly with temperature. For instance, it is surprising to observe that R_{HS} as well as $R_{r/g}$ first decreases with the elevated temperature ranging between

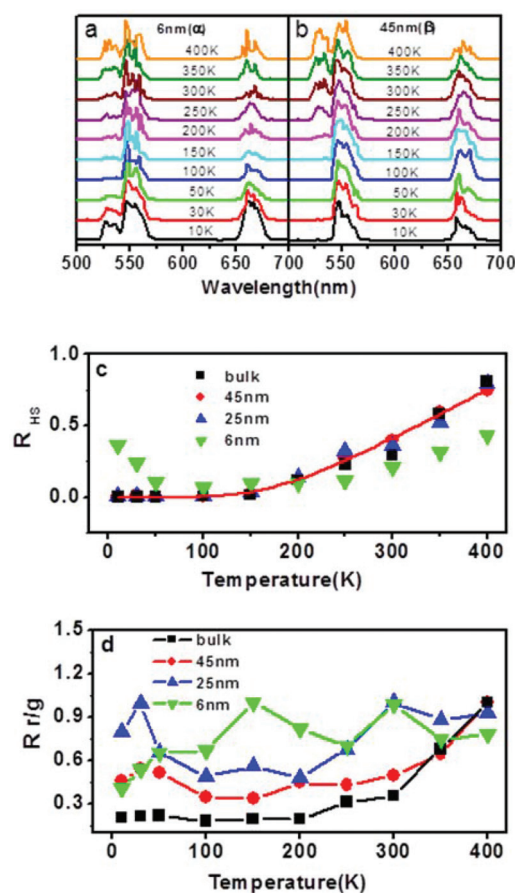


Fig. 5 The UCL spectra of NaYF₄:Yb³⁺, Er³⁺ nanoparticles of 6 nm (a) and 45 nm (b) when excited at 980 nm at verified temperatures from 10 K to 400 K; the intensity ratio of $^2H_{11/2}-^4I_{15/2}$ to $^4S_{3/2}-^4I_{15/2}$ (c). The intensity ratio of red to green (d).

10–100 K, and it then increases with the elevated temperature in the range of 150–300 K.

For comparison, the temperature-dependent UCL spectra of the 45 nm β -phase NaYF₄ sample are shown as Fig. 5(b). It can be seen that the 45 nm and the 6 nm samples demonstrate different temperature-dependent UCL behavior. The obvious differences can be concluded as: (1) the Stark splitting of the emission lines for Er³⁺ ions can be distinguished at any temperature. (2) R_{HS} increases solely with the elevated temperature. When the temperature is below 100 K, the transition of $^2H_{11/2}-^4I_{15/2}$ cannot be identified due to low electron population of $^2H_{11/2}$. It should be highlighted that for the other samples, (25 nm, bulk), the temperature-dependent behavior of the UCL is similar to that of the 45 nm sample. In order to better understand the temperature-dependent behavior in different samples, R_{HS} and $R_{r/g}$ as a function of temperature in different samples are shown as Fig. 5(c) and (d), respectively. As is well known, the intensity ratio R_{HS} of $^2H_{11/2}-^4I_{15/2}$ to $^4S_{3/2}-^4I_{15/2}$ is sensitive to temperature, and is a critical parameter to mark the temperature change in UCL processes. In Fig. 5(c), R_{HS} for the β -phase (25, 45 nm and bulk) NaYF₄ UCNCs increases solely with elevated temperature. The

population distribution on $^2\text{H}_{11/2}$ and $^4\text{S}_{3/2}$ should be dominated by Boltzmann's thermal distribution and elevated temperature leads to rapid population of $^4\text{S}_{3/2}$ – $^2\text{H}_{11/2}$. The population ratio of $^2\text{H}_{11/2}$ – $^4\text{I}_{15/2}$ to $^4\text{S}_{3/2}$ – $^4\text{I}_{15/2}$ can be written as,

$$R_{\text{HS}} = \alpha e^{-\Delta E/KT} \quad (1)$$

where α is a constant, ΔE is the energy difference between the two levels ($^2\text{H}_{11/2}$ and $^4\text{S}_{3/2}$), K is Boltzmann's constant, and T is the absolute temperature. The experimental data of the β -phase samples can be fitted by eqn (1) very well. The 6 nm α -phase NaYF₄ sample differs from the others, in which the R_{HS} decreases with the increasing temperature and reaches a maximum around 100 K. Until now, this behavior has not been completely understood. Previously, Liu and co-workers studied the temperature-dependence of photoluminescence of Er³⁺ ions in Y₂O₃ nanotubes and also observed an indeterminate phenomenon. They observed that even at low temperature (~10 K), the population of $^2\text{H}_{11/2}$ could not be neglected and the photoluminescence of $^2\text{H}_{11/2}$ – $^4\text{I}_{15/2}$ could be observed, unlike the micro-sized powders and they attributed it to the contribution of special electron–photon coupling modes for one-dimensional NPs.³² In the present case, we suggest that the indeterminate temperature-dependence of R_{HS} for the 6 nm sample is related to its imperfect crystallinity. In the disordered system of the local environment, the nonradiative transition rate of $^2\text{H}_{11/2}$ – $^4\text{S}_{3/2}$ might be very large and might be comparable to the radiative transition rate of $^2\text{H}_{11/2}$ – $^4\text{I}_{15/2}$ and even the thermal activation rate of $^4\text{S}_{3/2}$ – $^2\text{H}_{11/2}$. In this case, R_{HS} can not be simply described by eqn (1), but should be described by a modified model.³⁵ It should be also pointed out that the indeterminate relationship between R_{HS} and temperature might be also related to the local thermal effect induced by the laser irradiation.

The intensity ratios $R_{\text{r/g}}$ as a function of temperature were also obtained, as shown in Fig. 5(d). It can be observed that for the bulk sample, $R_{\text{r/g}}$ increases solely with the elevated temperature, which can be attributed to the increase of non-radiative relaxation rates of $^4\text{S}_{3/2}$ – $^4\text{F}_{9/2}$, $^4\text{I}_{11/2}$ – $^4\text{I}_{13/2}$. For the other samples, the variation tendency is similar above 100 K, however, the variation degree is not as large as the bulk sample, and the variation below 100 K is more complex.

3.4 Temperature-dependent UCL dynamics

In order to deeply understand the UCL behavior in NaYF₄:Yb³⁺, Er³⁺ UCNCs with different sizes and phases, the temperature-dependent dynamical processes of the excited states ($^4\text{F}_{13/2}$, $^4\text{S}_{3/2}$, $^4\text{F}_{9/2}$) for Er³⁺ ions were measured and compared under the excitation of a 980 nm pulsed laser. Fig. 6 shows the typical dynamical processes of 6 nm (α -phase) and 45 nm (β -phase) NaYF₄:Yb³⁺, Er³⁺ at different temperatures. It can be seen that in the 6 nm α -phase sample no rising process was observed for all the $^2\text{H}_{9/2}$ – $^4\text{I}_{15/2}$, $^4\text{S}_{3/2}$ – $^4\text{I}_{15/2}$ and $^4\text{F}_{9/2}$ – $^4\text{I}_{15/2}$ transitions (see Fig. 6(a–c)). An immediate decay of the emission intensity after the excitation pulse (without a rise) points

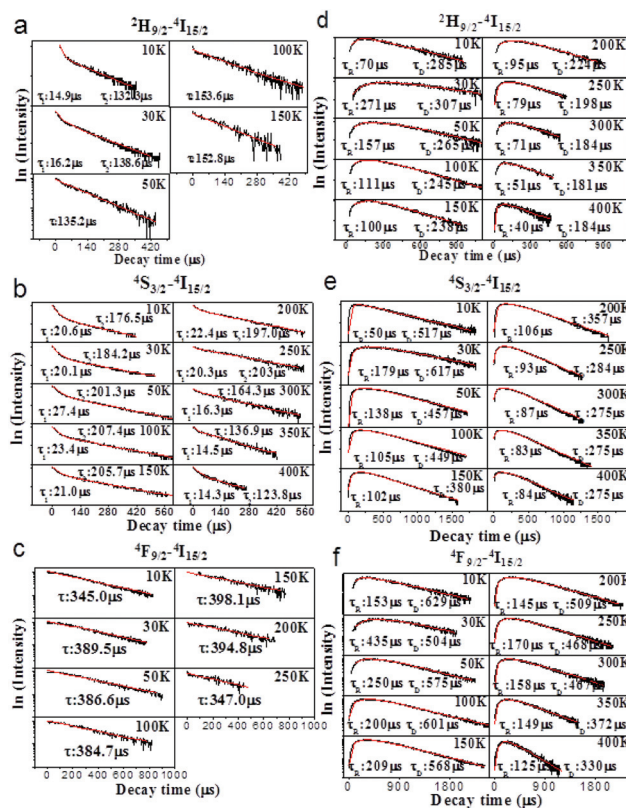


Fig. 6 The UCL decay time of the transitions at 416, 544 and 656 nm of Er³⁺ ions in 6 nm α -NaYF₄ (a), (b) and (c); 45 nm β -NaYF₄ (d), (e) and (f) samples with different temperature.

toward ground state absorption followed by excited state absorption (GSA/ESA).³³

This result suggests that as the particle size is small enough, the ET from Yb³⁺ to Er³⁺ is insufficient. There are several possible factors leading to this point. First of all, as the particle size decreases to several nanometers, the ET efficiency will be suppressed due to the boundary effect of nanoparticles.³⁴ Second, in the α -phase UCNCs prepared at relatively low temperature, a number of defect states should be involved, which can be judged from imperfect crystallinity of the sample. Furthermore, a number of large phonon bonds such as OH[−] groups would be involved unavoidably on the surface of the nanoparticles.³² These two facts will result in the improvement of nonradiative ET processes from Yb³⁺ ions to defect states and/or OH[−] groups, which compete with the ET processes from Yb³⁺ to Er³⁺. From Fig. 6(a–c) it can be also seen that for the $^2\text{H}_{9/2}$ – $^4\text{I}_{15/2}$ and $^4\text{S}_{3/2}$ – $^4\text{I}_{15/2}$ transitions, the decays of the emission intensities are all bi-exponential, which depend strongly on temperature and can be written as,³⁶

$$I(t) = I_1 \exp(-t/\tau_1) + I_2 \exp(-t/\tau_2), \quad (2)$$

where τ_1 and τ_2 represent the shorter and longer lifetime constants, respectively, and I_1 and I_2 are the original contribution for the shorter and longer decay times, respectively. The shorter time constants are in the range of several tens of

microseconds, while the longer time constants and in the range of several hundreds of microseconds. From the temperature-dependent dynamics it can be seen that for the $^2\text{H}_{9/2}-^4\text{I}_{15/2}$ transition, the contribution of the shorter decay process decreases quickly with the increasing temperature, and when the temperature is above 30 K, the shorter decay process nearly disappears. For the $^4\text{S}_{3/2}-^4\text{I}_{15/2}$ transition, the relative contribution of the shorter decay process slowly decreases with the elevated temperature. It is suggested that the appearance of bi-exponential decay processes comes from the emissions of Er^{3+} ions at different local environments, the shorter process corresponds to the emissions of Er^{3+} ions with disordered local environment or on the surface of nanoparticles, while the longer process corresponds to the emissions of Er^{3+} ions at relatively neat crystalline local environment. It should be noted that in some of our previous works, we studied the site symmetry of some oxide compounds, such as $\text{Y}_2\text{O}_3:\text{Eu}^{3+}$ and $\text{YVO}_4:\text{Eu}^{3+}$ nanowires and nanoparticles by laser-selective excitation and observed that as the particle size decreases to several nanometers, the emissions of Eu^{3+} ions originating from surface sites occur, which are with relatively broad lines and short lifetimes.^{37,38}

The UCL of Er^{3+} ions with different local environments demonstrate different temperature-dependent behavior. The emissions of Er^{3+} ions with disordered local environments demonstrate quick temperature quenching, while Er^{3+} ions with neat crystalline local environments yield slow temperature quenching. This is well in accordance with the temperature dependence of emission spectra.

The temporal evolution of β -phase $\text{NaYF}_4:\text{Yb}^{3+}, \text{Er}^{3+}$ (45 nm) as a function of time can be fitted to a Vial's type equation, for all the $^2\text{H}_{9/2}-^4\text{I}_{15/2}$, $^4\text{S}_{3/2}-^4\text{I}_{15/2}$ and $^4\text{F}_{9/2}-^4\text{I}_{15/2}$ transitions:

$$I(t) = I_1 \exp(-t/\tau_D) - I_2 \exp(-t/\tau_R) \quad (3)$$

where τ_D and τ_R represent the decay and the rise of the emission, respectively. τ_D is essentially the UCL lifetime, which is influenced by the lifetime of the emitting level and the lifetime of the levels feeding the excited states through the energy transfer process. The rise time τ_R is determined by the lifetime of the emitting state and the ET rate from Yb^{3+} to Er^{3+} .^{37,39} From the best fitting functions it can be seen that the rising time constant first increases with the increase of temperature when the temperature varies from 10 K to 30 K, and it then decreases with the further increase of temperature. It should be noted that for the 25 nm and bulk samples, their luminescent dynamics are similar to those of the 45 nm samples, and their temperature-dependent dynamics are shown as Fig. S1 and S2,[†] respectively.

In order to better understand the temperature-dependent luminescent dynamics, the rising and decay time constants as a function of temperature for different samples are shown as Fig. 7. It can be seen that the rising time constants for the 25 nm, 45 nm and the bulk samples demonstrate similar variation tendency. Generally, in the 25 nm, 45 nm and the bulk samples, for all the transitions the rising time constants first

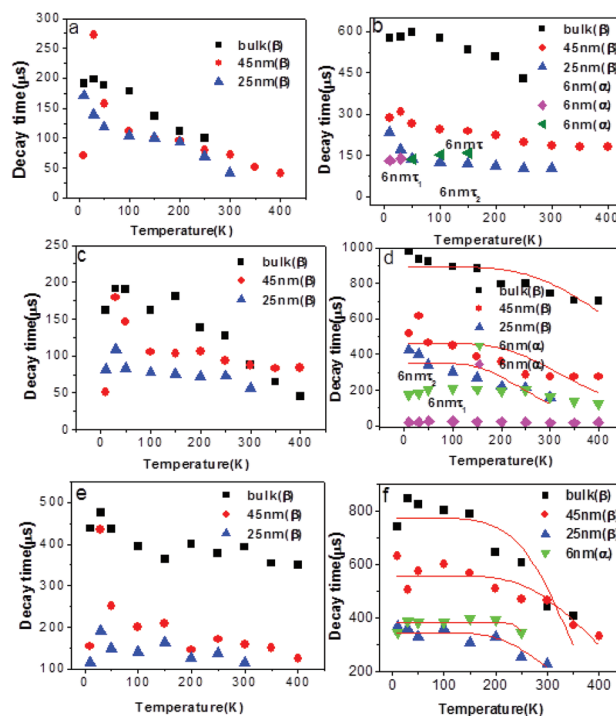


Fig. 7 Decay time constant versus temperature of $\text{NaYF}_4:\text{Yb}^{3+}, \text{Er}^{3+}$ NCs with different size. The rising time of $^2\text{H}_{9/2}-^4\text{I}_{15/2}$ (a); decay time of $^2\text{H}_{9/2}-^4\text{I}_{15/2}$ (b); rising time of $^4\text{S}_{3/2}-^4\text{I}_{15/2}$ (c); decay time of $^4\text{S}_{3/2}-^4\text{I}_{15/2}$ (d); rising time of $^4\text{F}_{9/2}-^4\text{I}_{15/2}$ (e); decay time of $^4\text{F}_{9/2}-^4\text{I}_{15/2}$ (f) transition of Er^{3+} ions with different sizes.

rapidly increase with the increase of temperature from 10 K to 30 K, and then gradually decrease with the increase of temperature above 30 K due to the increase of nonradiative relaxation rates for different emission levels (see Fig. 7(a), (c), (e)). From Fig. 7(b), (d) and (f), it can be seen that for most cases, the decay time constants decrease with the increase of temperature. As is well known, the decay time constant is the reverse of the sum of the radiative and nonradiative transition rates of the cross-relaxation processes. The radiative decay rate for an emission level rarely changes with temperature, while the nonradiative decay rate varies significantly with temperature, thus the luminescent lifetime τ for any level can be roughly expressed as,

$$\tau = \frac{1}{W + W(0)[1 - \exp(-\hbar\omega/kT)]^{-\Delta E/\hbar\omega}} \quad (4)$$

where W represents the radiative transition rate for the emission levels, $W(0)$ represents the nonradiative transition rate from the emission level to the nearest down level at 0 K, $\hbar\omega$ is the phonon energy for multi-phonon relaxation process and ΔE is the energy gap between the emission level and the nearest down level. The majority of experimental data can be well fitted by eqn (4). By fitting, W and $W(0)$ for different emission levels $^4\text{S}_{3/2}$ and $^4\text{F}_{9/2}$ were obtained, and the optimum fitting values were listed in Table 1. It can be seen that W increases with the decrease of nanocrystalline particle size. There are two factors affecting the radiative transition rate of

Table 1 A list of W and $W(0)$ for different emission levels in $\text{NaYF}_4\text{:Yb}^{3+}$, Er^{3+} samples with different sizes: bulk, 45 nm, 25 nm and 6 nm

	Samples	Bulk	45 nm	25 nm	6 nm
$^4\text{S}_{3/2}-^4\text{I}_{15/2}$	W/ms^{-1}	1.01	1.25	1.59	5.03
	$W(0)/\text{ms}^{-1}$	0.08	0.75	1.23	
$^4\text{F}_{9/2}-^4\text{I}_{15/2}$	W/ms^{-1}	1.24	1.78	2.4	2.43
	$W(0)/\text{ms}^{-1}$	0.90	0.50	0.47	0.18

RE ions doped NCs, effective refractive index effect and lattice distortion effect.⁴⁰ The radiative rate for the electric-dipole transition increases with the increase of refractive index of the phosphor. Meltzer *et al.*⁴¹ reported that the radiative transition rate of $\text{Y}_2\text{O}_3\text{:Eu}$ NCs is dependent of not only the refractive index itself, but also the surrounding medium, which can be expressed as, $n_{\text{eff}} = xn + (1 - n)n_{\text{med}}$, where x is the filling factor showing what fraction of space is occupied by the nanocrystals and n_{med} is the refractive index of medium. Based on this function, the radiative transition rate for RE ions should decrease in comparison to the corresponding bulk material owing to the effective refractive index effect.⁴¹ However, in some NCs, the imperfect crystallinity could induce the degeneration of lattice symmetry, and thus the radiative transition rate increases with the decrease of particle size.⁴² In Table 1, it can be also observed that $W(0)$ increases with the decrease of particle size for the $^4\text{S}_{3/2}-^4\text{I}_{15/2}$ transition, however, for the $^4\text{F}_{9/2}-^4\text{I}_{15/2}$ transition, $W(0)$ surprisingly increases with the decrease of particle size, which has not been clearly understood yet.

4. Conclusions

In this work, we studied the temperature dependence of UCL and dynamics of $\text{NaYF}_4\text{:Yb}^{3+}$, Er^{3+} UCNCs with different particle sizes and crystalline phases and observed some interesting phenomena. First of all, we observed that for the 6 nm α -phase $\text{NaYF}_4\text{:Yb}^{3+}$, Er^{3+} UCNCs, the overall UCL intensity decreased solely with the increasing temperature, while for the β -phase $\text{NaYF}_4\text{:Yb}^{3+}$, Er^{3+} UCNCs ranging of several tens to thousands of nanometers, the UCL intensity as a function of temperature demonstrated an optimum around 100 K. Second, for the β -phase $\text{NaYF}_4\text{:Yb}^{3+}$, Er^{3+} UCNCs the intensity ratio of $^2\text{H}_{11/2}-^4\text{I}_{15/2}$ to $^4\text{S}_{3/2}-^4\text{I}_{15/2}$ (R_{HS}) increased solely with temperature and could be well fitted by the Boltzmann's distribution function, while for the α -phase $\text{NaYF}_4\text{:Yb}^{3+}$, Er^{3+} UCNCs R_{HS} demonstrated a complex and obscure temperature dependence. Third for the β -phase $\text{NaYF}_4\text{:Yb}^{3+}$, Er^{3+} UCNCs, the UC dynamics demonstrated a rising process for all the visible UC transitions due to the sufficient ET from Yb^{3+} to Er^{3+} , while for the α -phase $\text{NaYF}_4\text{:Yb}^{3+}$, Er^{3+} UCNCs, the rising process completely disappeared due to insufficient ET, and the decay processes were bi-exponential due to surface effects. Their physical origins were carefully discussed.

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